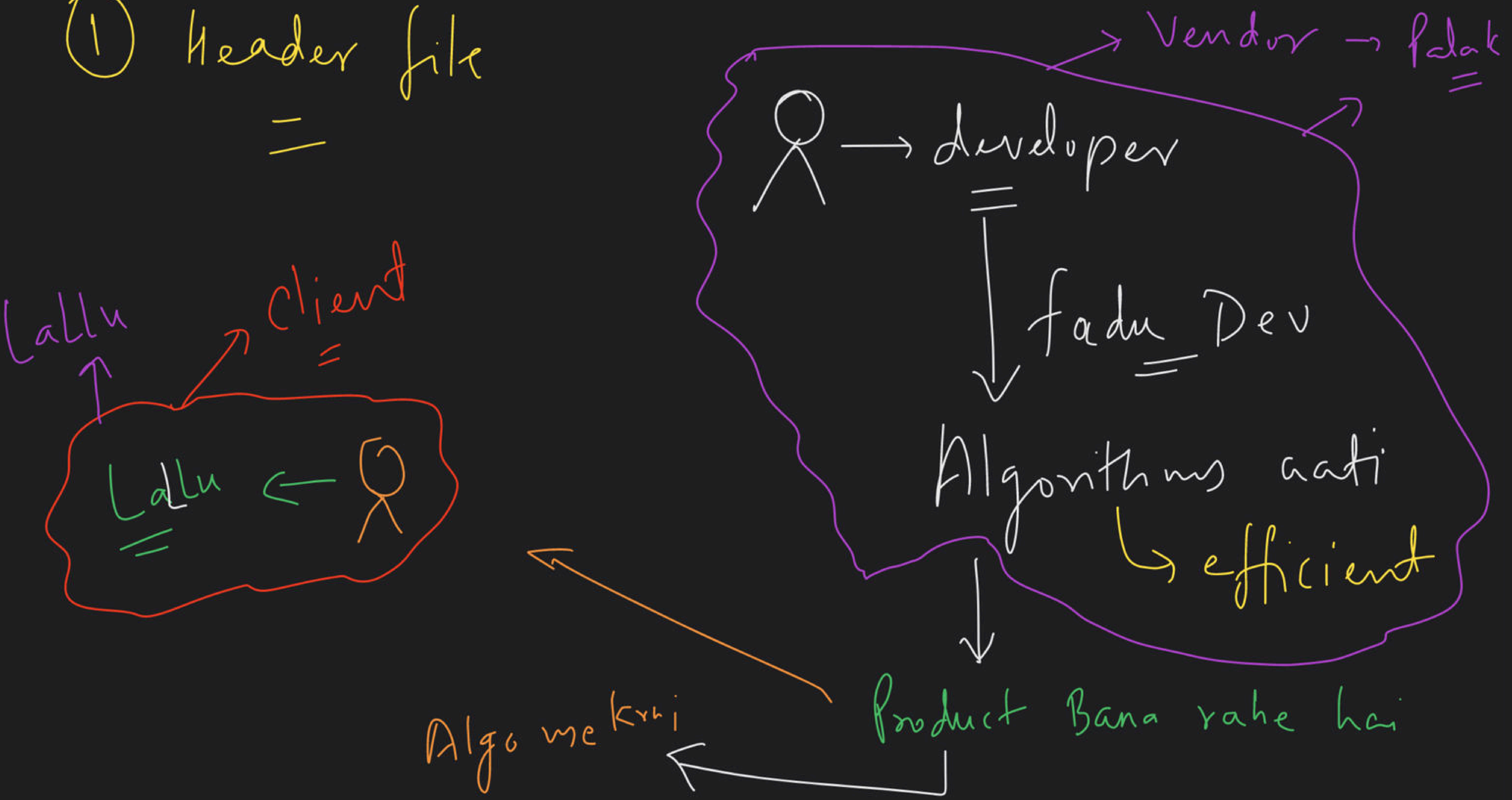
Mega Class: Stacks

Special class

Love Babbar • Aug 11, 2024

1 Header file



Palak → vendor

Algo

share w/o giving impl

header files + compiled

↓

object file

Signature

.cpp

.h

.o

linker

.exe

2 Usecase

Siya , Mukesh

Single Team → Product

Mukesh

↓

feature A

featureB

help.cpp

call911( )

5

~~Sharing.cp~~

me

help.h me

call911(.....)

help.h

.o

1

1. Reusability
2. Code sharing

→ DRY

3. Modularity
4. Code Org.

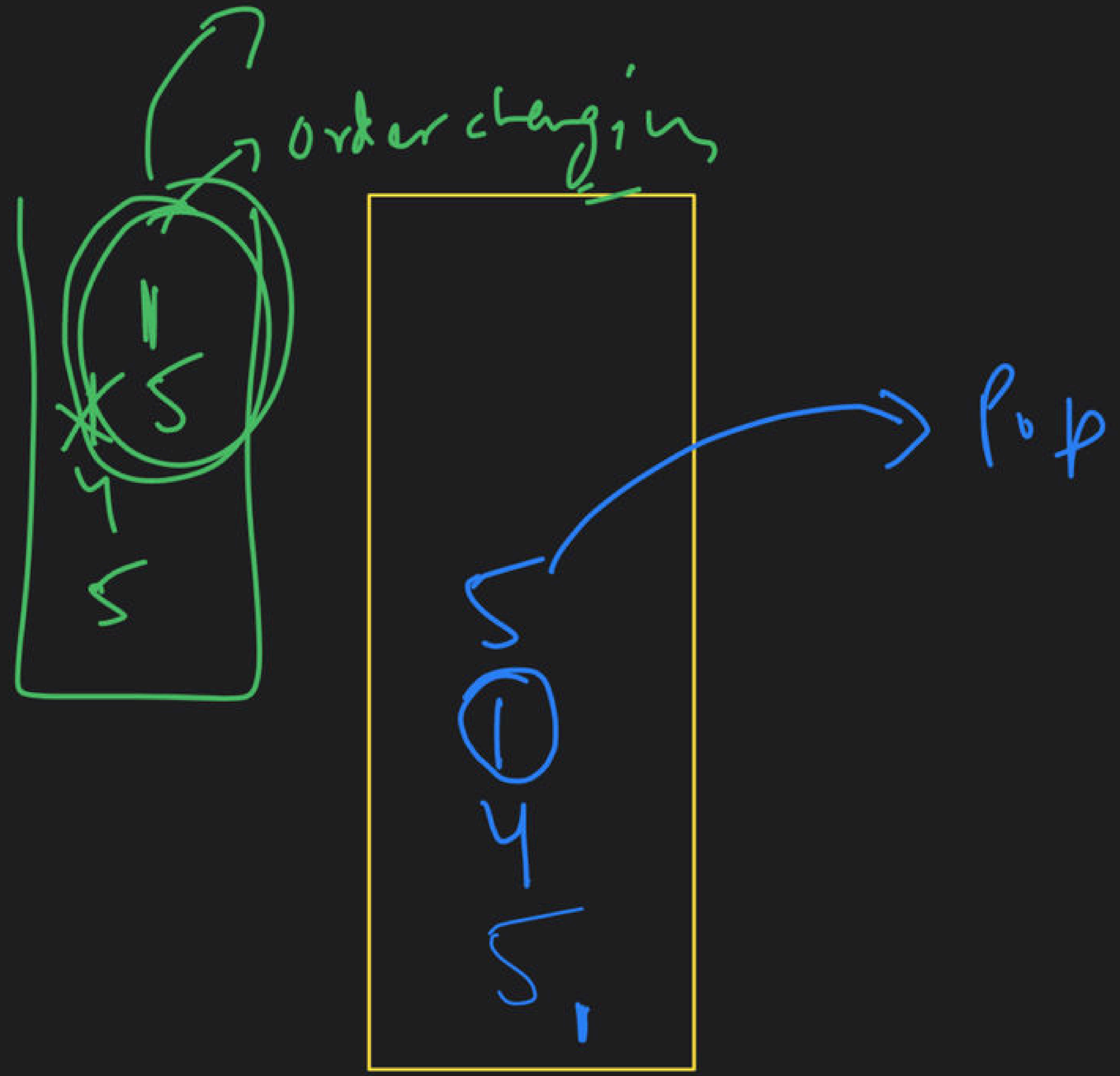
1 Min Stack [leetcode : 155]

5 min → Read & Think

Back : 3:05

5, 4, 1, 5, 2, 4, 2, 5, 6,

O(1)



Push,

Pop → RE

→ It

→ O(n)

5, 4, 1, 1

Push

Pop

Top → ?

getmin() → ✓✓

5, 4, 1, 5, 2, 4, 2, 0, 5, 6

~~6, 0~~

~~5, 0~~

~~0, 0~~

2, 1

4, 1

2, 1

5, 1

1, 1

4, 4

5, 5

same

we pair

↳ store Both

{value, min. Tillnow}

6 → Push → Top min till now ? ?

0

→ Top se → O(1)

② LC: 907

"Sum of Subarray Minimums"

7 min to Think & Read

⇒ 3:30 PM

arr = [3, 1, 2, 4]

$\quad\;\;$ 0 $\;$ 1 $\;$ 2 $\;$ 3

$O(n^2)$

[3] = 3

[3 1] = 1

[3 1 2] = 1

[3 1 2 4] = 1

[1] = 1

[1 2] = 1

[1 2 4] = 1

[2] = 2

[2 4] = 2

[4] = 4

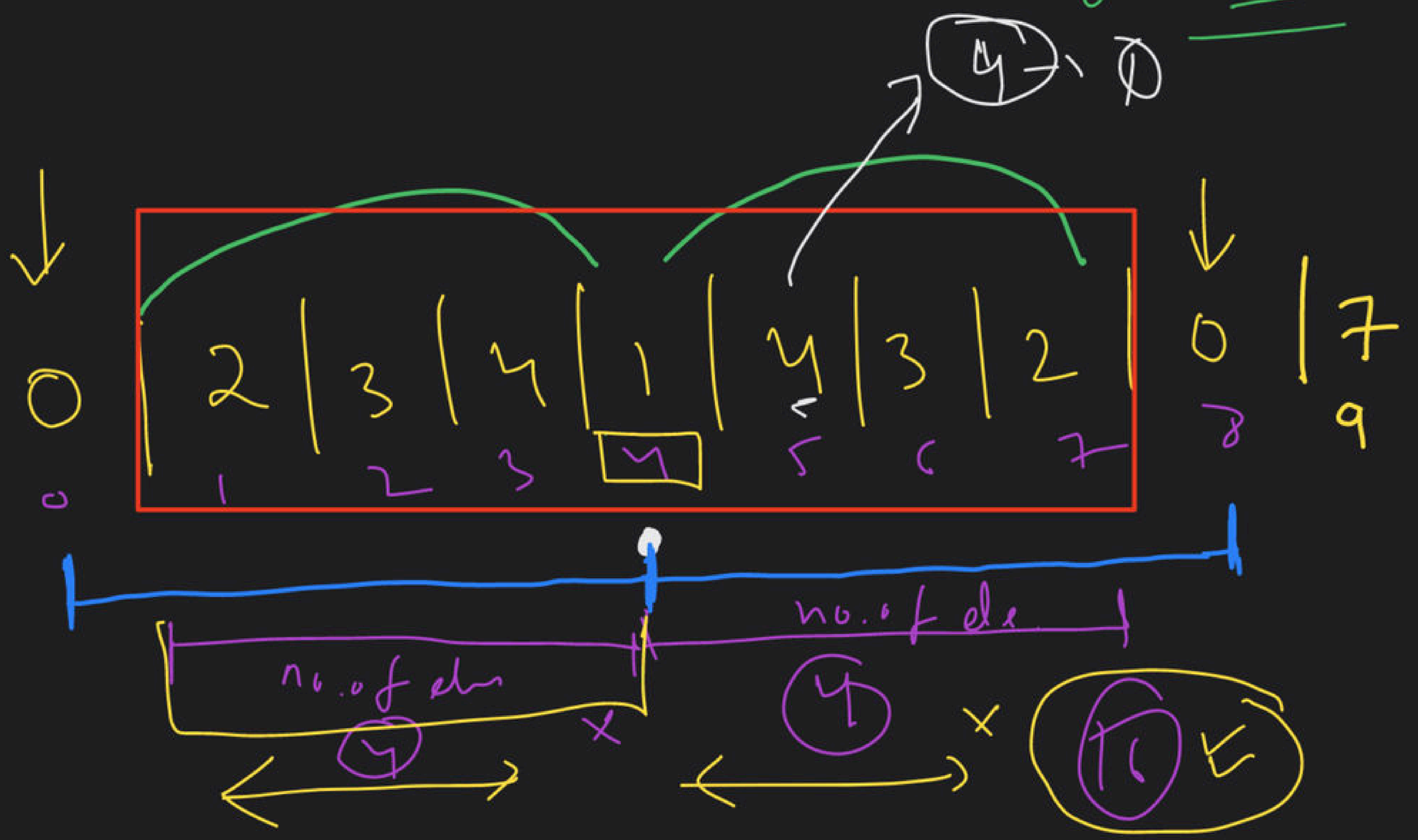
1 time $\quad$ 6 times $\quad$ 2 times $\quad$ 1 time

$3 + 1 + 1 + 1 + 1 + 1 + 1 + 2 + 2 + 4 = 17$ return

M1 ⟹ Brute force ⟹ find all subarray & get min of all

M2

How many times a particular element is included into my ans?



4 1 4 3
4 1 4 3 2
4 1 4
3 4 1 4 3
3 4 1 4 3 2
2 3 4 1
3 4 1
4 1
1
1 4
1 4 3
1 4 3 2
2 3 4 1 4
2 3 4 1 4 3
2 3 4 1 4 3 2
3 4 1 4

2 | 3 | 4 | 1 | 4 | 3 | 2

0 1 2 3 4 5 6 7 8

4 × 4 = 16

4 × 4

2
2 3
2 3 4
2 3 4 1
2 3 4 1 4
2 3 4 1 4 3
2 3 4 1 4 3 2

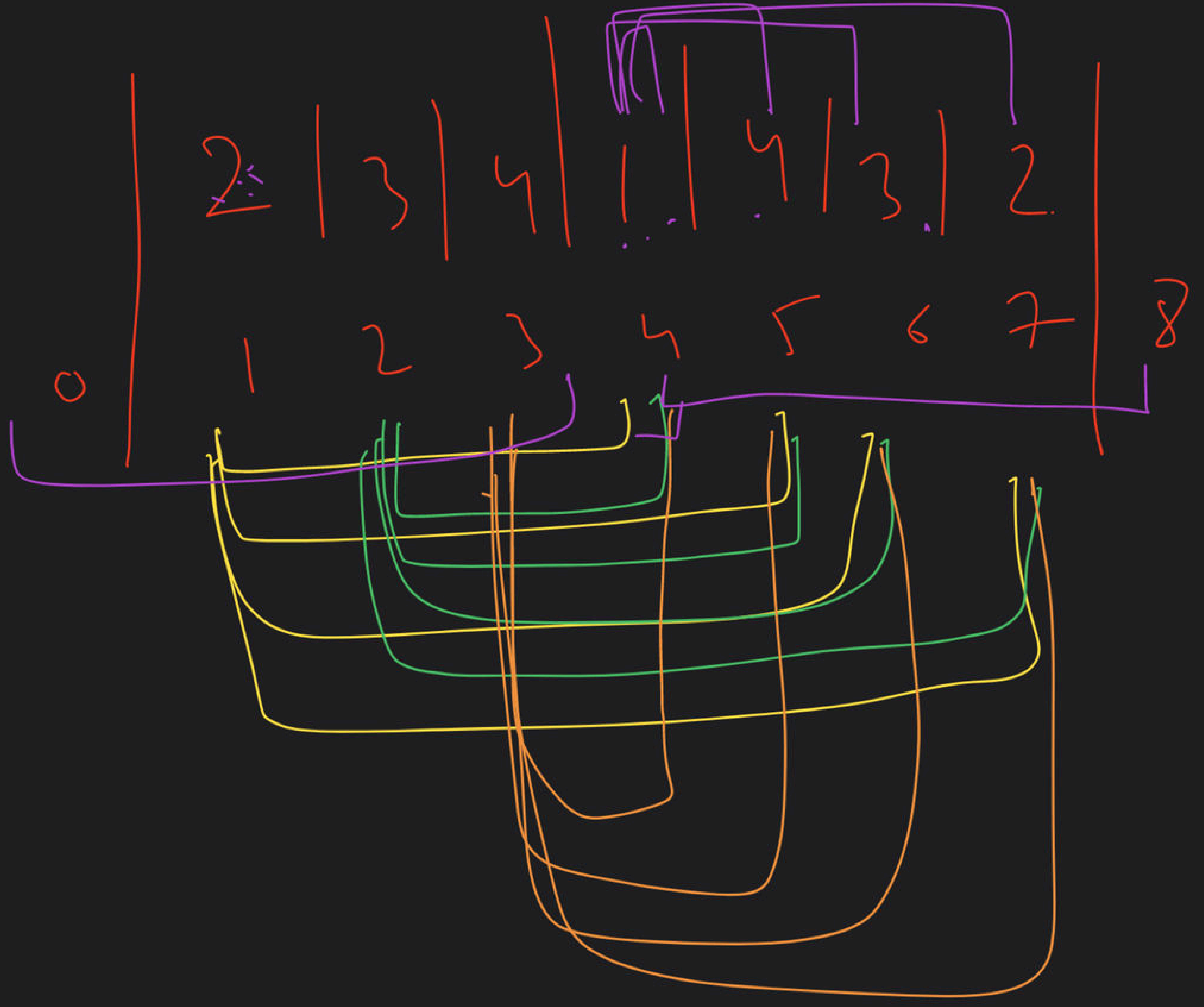
16

3 4 1
3 4 1 4
3 4 1 4 3
3 4 1 4 3 2

4 1
4 1 4
4 1 4 3
4 1 4 3 2

1
1 4
1 4 3
1 4 3 2

4
4 3
4 2



16x1

16

4

4 x4

16

-1 3 1 2 4 4

0 1 2 3

2 3 → $2 \times 3 = 6$

6

3 1

3 1 2

3 1 2 4

1

1 2

1 2 4

→ 6

nextsmallindex - i ⇒

4 - 1 = 3

i - prevsmallind

1 - (-1) = 2

6

$8, 15, 7 \longrightarrow [15]$ ✓

$\;0 \quad 1 \quad 2$

NSI : 2

PSI : 0

$1 \times 1 = 1$

Left = $1 - 0 = 1$

Right = $2 - 1 = 1$

1

1

6	2	3	4	1	4	3	2	0
0	1	2	3	4	5	6	7	8

$\Rightarrow$

$$(NSI - i) * [i - PSI]$$

$$1 * 1 = 1$$

$[4] \Rightarrow$

Algo

1. for every index

1.1 Prev smaller Index
next " "

1.2 Left level = $i - PSI$
Right = $NSI - i$

1.3 Sum = Sum + (Left × Right)

③ LC : 2104

"Sum of Subarray Ranges"

4:40 → Read & Think

4 - 1 = 1

4	-2	-3	4	1
0	1	2	3	4

-2+2 = 0

4 → {4, 4} ⇒ 0

4 -2 → {4, -2} ⇒ 6

4 -2 -3 → {4, -3} ⇒ 7

4 -2 -3, 4 → {4, -3} ⇒ 7

4 -2 -3 4 1 → {4, -3} ⇒ 7

-2 → {-2, -2} ⇒

-2 -3 → {-2, -3}

-2 -3 4 → {4, -3}

-2 -3 4 1 → {4, -3}

-3 - -

-3 4 -

-3 4 1 -

4 -   1 -

4 1 -   -

TotalMaxSum ⇒ Σ sub → max

" Min " ⇒ Σ sub → min

(TMS - TMins) → Return

[1, 3, 2]

find Range of each Subarr & add them Return

1 => 1 - 1 = 0

1 3 => 3 - 1 = 2

1 3 2 => 3 - 1 = 2

3 => 3 - 3 = 0

3 2 => 3 - 2 = 1

2 => 2 - 2 = 0

0 + 2 + 2 + 0 + 1 + 0 = 5

Maximum of all subarr => 1 + 3 + 3 + 3 + 3 + 2 => 15

Mini of " " => 1 + 1 + 1 + 3 + 2 + 2 = 10

15 - 10

M2 intuition

LC 907

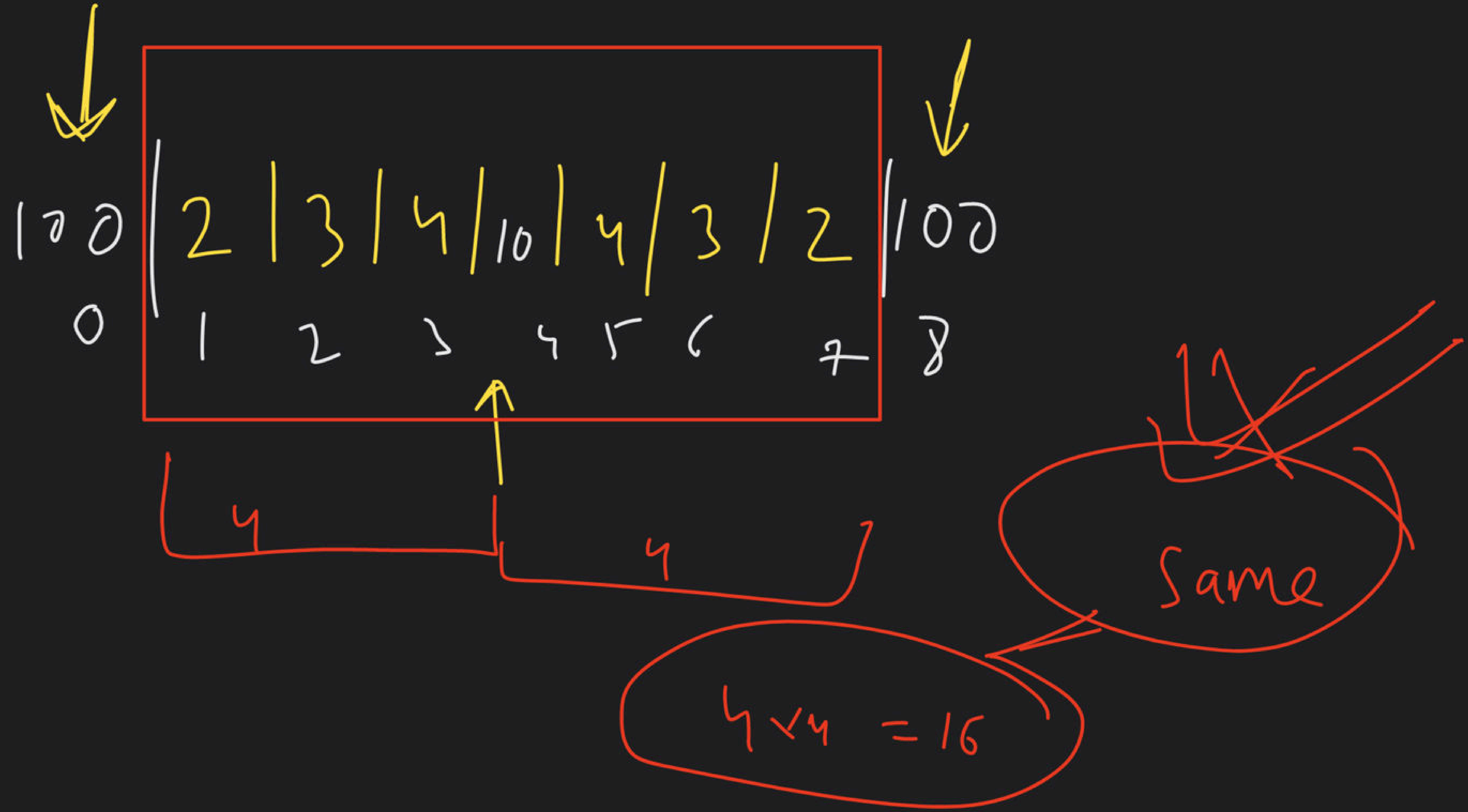
1 Sum of Maximums of each subarray

2 " " Mini " " "

3 ANS = 1 - 2

STEP → How can I find 1 & 2

in O(n)



100 | 2 | 3 | 4 | 10 | 4 | 3 | 2 | 100
0 1 2 3 4 5 6 7 8
4
4
Same
4×4 = 16

4 LC 224: Basic Calculator

5 min $\Rightarrow$ Think & Read

5:10 PM

" 1 + (2 − (3 + 4)) + 5 "

Stack

Result ⇒ keep track of the result & sign

(→ maintain

) → Pop combine ans to result

$1 + (2 - (3 + 4)) + 5$

$\rightarrow -5 + 1$

result = $0 \ldots -5$

num = $0 \ldots 5$

sign = $+ \ldots +$

$-7 + 2 \Rightarrow -5$

7

$2 \quad -7$

$-$

2

int =

$= -5$

$-5 + 1$

$+ \quad = -4$

1

-4 ans

$3 + (+ \times 4)$

$1 \rightarrow$ ans

$7 \Rightarrow -7$

result = result + num $\times$ sign

$-4+5$

```
if ( digit )
{
num =   num   +  (c - '0')  → int

}
else if (  c = = '+' )
 {   result = result +  sign * num ;
       sign = 1
       num = 0
 }
```

(c == '-')

result = result + (sign * num)

sign = -1
num = 0

'(' →

Result → Push
sign ⇒ Push → (+1/-1)
result = 0
sign = 1

')'

result = sign * num;

ch ~~Tak~~ k

// Pop sign → Pop kuk

result *= sign

result += st.top();

num = 0;

" 12345 "

ch = 1

num = 1

ch = 2

L: num = num * 10 + ch

LC = 224 ⇒ Dry. Run khudse

$0 + (+50 - 1 - (+4 + 5 + 2) - 3) + (+6 + 8)$

result = 14

num = 0

sign = 1

$35 + 14$

$\Rightarrow 49$

ans

$49 - 11$

49

35